

Number Systems - Assignments.

Q1] Complete the table with equivalents in Binary, Octal and Hexa for the following.

$$(143.75)_{10} = (?)_2$$

$$143 = 1001111$$

$$2 \overline{) 143}$$

$$2 \overline{) 71} - 1$$

$$2 \overline{) 35} - 1$$

$$2 \overline{) 17} - 1$$

$$2 \overline{) 8} - 1$$

$$2 \overline{) 4} - 0$$

$$2 \overline{) 2} - 0$$

$$1$$

$$0.75 = 0.11$$

$$0.75 \times 2 = 1.5 - 1$$

$$0.5 \times 2 = 1.0 - 1$$

$$0.75 \times 8 = 6.0$$

$$(143.75)_{10} = 1001111.11_2$$

$$8 \overline{) 143}$$

$$8 \overline{) 17} - 7$$

$$8 \overline{) 2} - 1$$

$$(1001111.11)_2 = (?)_8$$

$$\begin{array}{ccccccc} 00 & 100 & 1111 & . & 110 \\ \hline 1 & 1 & 7 & . & 6 \end{array}$$

$$143.75_{10} = 217.6_8$$

$$1001111.11_2 = 117.6_8$$

$$\text{Hexa } 1001111.11_2 = (?)_{16}$$

$$16 \overline{) 143}$$

$$8 \overline{) 8} - F$$

$$8F.C_{16}$$

$$0.75 \times 16 = 12.0$$

$$\begin{array}{ccccccc} 01 & 00 & 1111 & . & 1100 \\ \hline 4 & F & C \end{array}$$

$$1001111.11_2 = 4F.C_{16}$$

$$11011 \cdot 101_2 = (?)_{10}$$

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$16 + 8 + 0 + 2 + 1 + 0.5 + 0.25 + 0.125$$

$$27.875$$

$$11011 \cdot 101_2 = 27.875_{10}$$

$$11011 \cdot 101_2 = (?)_8$$

$$\underbrace{01}_3 \underbrace{1011}_3 \cdot \underbrace{101}_5$$

$$11011 \cdot 101_2 = 38.5_{(8)}$$

$$11011 \cdot 101_2 = (?)_{16}$$

$$\underbrace{0001}_1 \underbrace{1011}_B \cdot \underbrace{1010}_A$$

$$11011 \cdot 101_2 = 1B \cdot A_{16}$$

$$453_8 = (?)_{10}$$

$$4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0$$

$$256 + 40 + 3$$

$$299$$

$$453_8 = 299_{10}$$

$$453_8 = (?)_2$$

$$453_8 = 635_2$$

$$453_8 = 100101011_2$$

$$\begin{array}{r} 8 \overline{) 453} \\ 8 \overline{) 51} \quad - 5 \\ \hline 6 \quad - 3 \end{array}$$

$$4538 = (?)_{16}$$

$$010001010011_{16}$$

$$4538 = 010001010011_{16}$$

$$\begin{array}{c} 01000 \\ \overbrace{0001}^1 \overbrace{0010}^2 \overbrace{1011}^B \end{array}$$

$$4538 = 12B_{16}$$

$$IF.C(16) = (?)_{10}$$

$$1 \times 16^1 + F \times 16^0 + C \times 16^{-1}$$

$$16 + 15 + 0.75$$

$$31.75$$

$$IF.C_{16} = 31.75_{10}$$

$$IF.C(16) = (?)_2$$

$$00011111.1100_{(2)}$$

$$IF.C(16) = (?)_8$$

$$\overbrace{00001111}^0 \cdot \overbrace{110000}^{60}$$

$$0 \quad 3 \quad 7 \quad \cdot \quad 6 \quad 0$$

$$IF.C_{16} = 37.6_8$$

Decimal	Binary	Octal	Hexa
143.75	1001111.11	217.6	8F.C
27.875	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	00011111.1100	37.6	IF.C

2) Find equivalents.

a) $324_5 = (?)_{10}$

$$3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0$$

$$75 + 10 + 4$$

$$\underline{89}_{10}$$

$$\underline{324_5 = 89_{10}}$$

b) $32.23_4 = (?)_{10}$

$$3 \times 4^1 + 2 \times 4^0 + 2 \times 4^{-1} + 3 \times 4^{-2}$$

$$12 + 2 + 0.5 + 0.1875$$

$$14.6875$$

$$\underline{32.23_4 = 14.6875_{10}}$$

c) $231_{10} = (?)_7$

$$\underline{231_{10} = 450_7}$$

$$\begin{array}{r} 7 \overline{) 231} \\ 7 \overline{) 33} \quad -0 \\ \underline{4} \quad -5 \end{array}$$

d) $189_{10} = (?)_6$

$$\underline{189_{10} = 513_6}$$

$$\begin{array}{r} 6 \overline{) 189} \\ 6 \overline{) 31} \quad -3 \\ \underline{5} \quad -1 \end{array}$$

3) Find the radix r

a) $(312/20)_r = (13.1)_r$

convert both sides to decimal.

$$312_r = 3r^2 + 1r + 2$$

$$= \frac{3r^2 + 1r + 2}{2r} \quad , \quad 20_r = 2r$$

$$13.1_r = r + 3 + \frac{1}{r}$$

$$\frac{3r^2 + 1r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$3r^2 + 1r + 2 = 2r^2 + 6r + 2$$

$$3r^2 - 2r^2 + 1r - 6r = 0$$

$$r^2 - 5r = 0$$

$$r(r-5) = 0$$

$$r-5 = 0$$

$$\boxed{r = 5}$$

b) $264r + 136r = 433r$

$$(2r^2 + 6r + 4) + (1r^2 + 3r + 6) = 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$3r^2 - 4r^2 + 9r - 3r + 10 - 3 = 0$$

$$-1r^2 + 6r + 7 = 0$$

$$r^2 - 6r - 7 = 0$$

$$r(r-6) - 7 = 0$$

$$(r-7)(r+1) = 0$$

$$\underline{\underline{r = 7}}$$

c) $221505_8 = 12345_r$

$$= 2(8^5) + 2(8^4) + 1(8^3) + 5(8^2) + 0(8) + 5$$

$$= 65536 + 8192 + 512 + 320 + 5$$

$$= 74565$$

$$12345_r = r^4 + 2r^3 + 3r^2 + 4r + 5$$

$$r = 16$$

$$16^4 + 2(16^3) + 3(16^2) + 4(16) + 5$$

$$= 65536 + 2192 + 768 + 64 + 5$$

$$= 74565$$

$$\underline{\underline{r = 16}}$$

4] Addition

a) $1101_2 + 11011111_2$

$$\begin{array}{r} 01 \\ 11011111 \\ \underline{111001100} \end{array}$$

$$1101_2 + 11011111_2 = \underline{\underline{111001100_2}}$$

b) $77652_8 + 76543_8$

$$\begin{array}{r} 77652 \\ 76543 \\ \underline{176415} \end{array}$$

$$77652_8 + 76543_8 = \underline{\underline{176415_8}}$$

c) $578A_{16} + ABDE_{16}$

$$A+E = 18_{16}$$

$$8+D+1 = 16_{16}$$

$$7+B+1 = 13_{16}$$

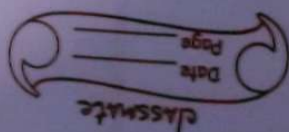
$$5+A+1 = 10_{16}$$

$$578A_{16} + ABDE_{16} = \underline{\underline{10388_{16}}}$$

d) $234_6 + 315_6$

$$\begin{array}{r} 234 \\ 315 \\ \underline{553} \end{array}$$

$$234_6 + 315_6 = \underline{\underline{553_6}}$$



5] Binary to Gray.

a) 10101_2
~~0000~~
1 0 1 0 1
1 1 1 1 1

$$10101_2 = 11111_2$$

b) 01110_2

0 1 1 1 0
~~0 0 0 0 0~~

$$01110_2 = 01001_2$$

6] Gray code to Binary.

a) 01101
0 1 1 0 1
0 1 0 0 1₂

b) 10101
1 0 1 0 1
1 1 0 0 1₂